Construction example

Instrument and available wood

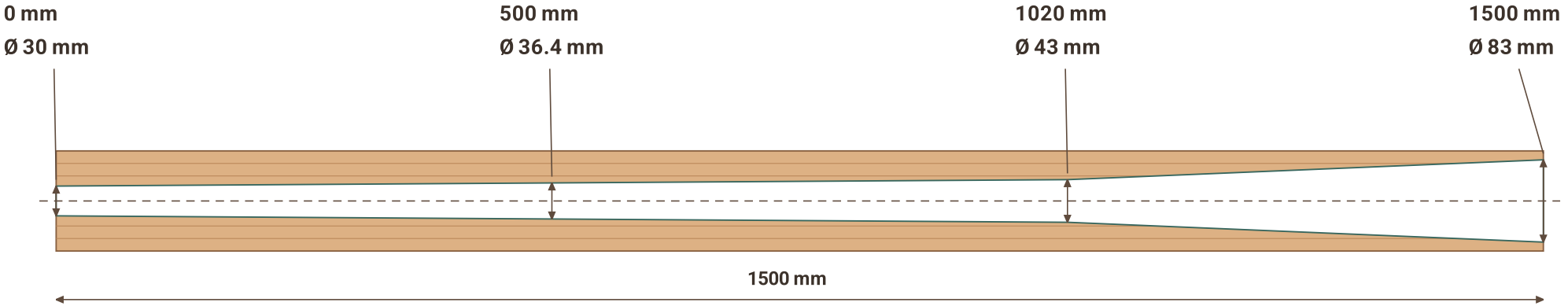
Instrument length: 1500 mm

Mouthpiece internal diameter: 30 mm

Bell internal diameter: 83 mm

Target note: C♯2 · 69.3 Hz

Available wood



White: final air passage · Longitudinal view · scale 1:6

Log outline: stated minimum diameter

Natural log

Length 1500 mm · Minimum outside diameter 101 mm

Minimum wall thickness: 9 mm

Geometric minimum: no additional allowance

Smallest remaining wall: 9 mm · at the bell

Wood condition and grain still need assessment. The target note is not a guarantee of the played note.

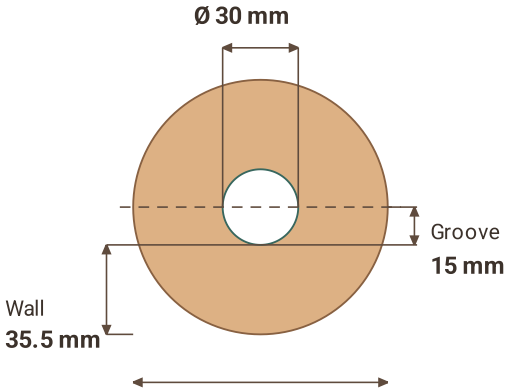
100 mm

100 mm - print at 100% and verify with a ruler

Construction example

Interior and carving dimensions

0 mm

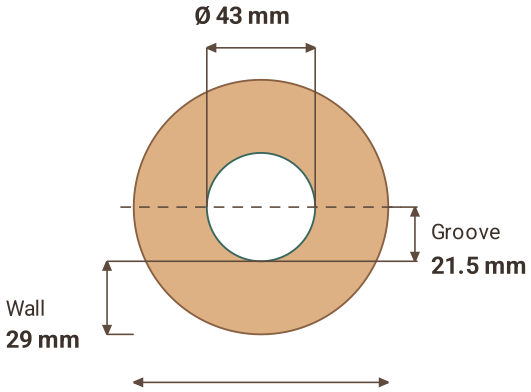


Width at joining plane 101 mm

Plane to outside (calculated) 50.5 mm

Cross-section scale 1:3

1020 mm

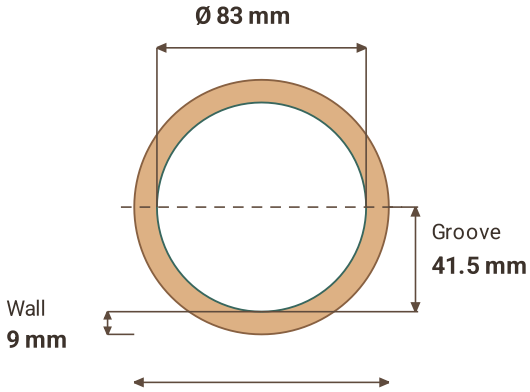


Width at joining plane 101 mm

Plane to outside (calculated) 50.5 mm

Cross-section scale 1:3

1500 mm



Width at joining plane 101 mm

Plane to outside (calculated) 50.5 mm

Cross-section scale 1:3

Carving measurements · millimetres

From the mouthpiece	Internal bore diameter	Groove depth per half	Remaining wall thickness
0	30	15	35.5
200	32.5	16.3	34.2
500	36.4	18.2	32.3
750	39.6	19.8	30.7
1020	43	21.5	29
1300	66.3	33.2	17.3
1500	**83**	**41.5**	**9**

Selected stations; complete measurements in CSV

Log: minimum wall estimated from the stated diameters, assuming a centred bore. Dimensions allow for the stated preparation losses.



100 mm - print at 100% and verify with a ruler